

HELLO!

I'm Sayeed Unnisa

I'm here, because I love giving presentations on **Java Full Stack**

JavaEE SERVLETS

Web Application:

It is a collection of web pages (Resources) used to provide service to end user.

Ex: web applications: [www.bookmyshow](http://www.bookmyshow.com), www.paytm.com, redbus, bigbasket, icici, net banking, youtube, gmail, facebook, timesofindia newspaper, twitter, amazon.
=>web application provides better services, saves time and faster service.

Webpages:

A webpage can be created using mainly HTML concept.

Types of web Pages:

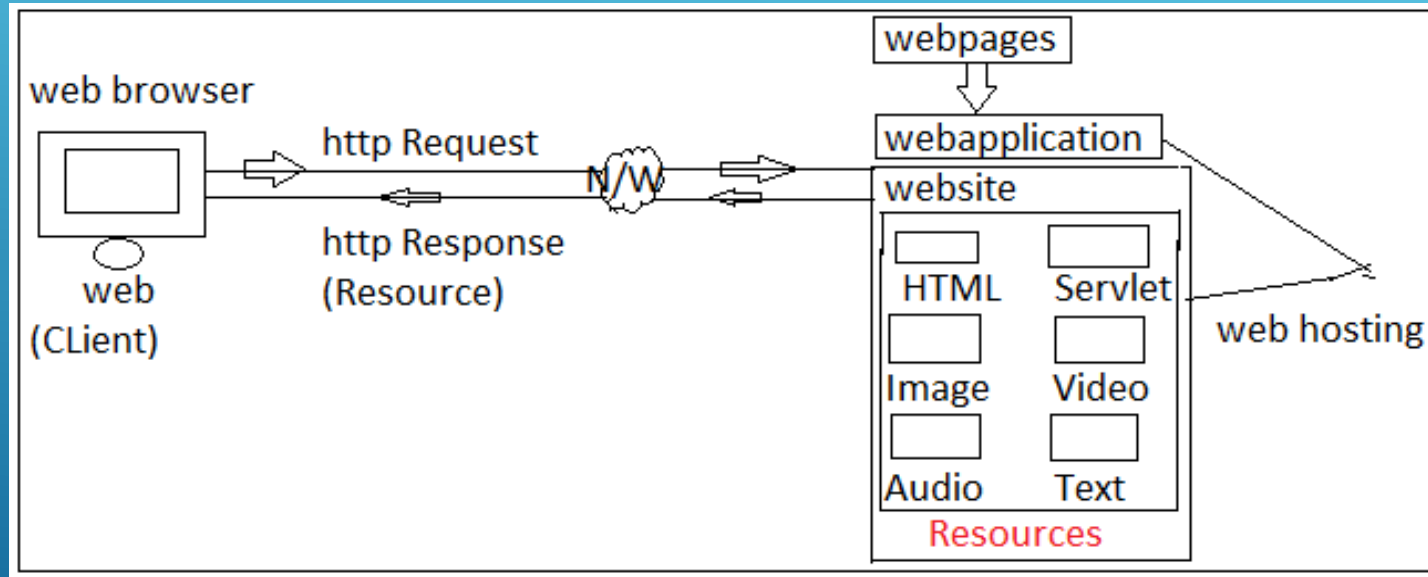
- 1. Static web page:** -- A page has fixed content show same to all end users, every time is known as static page.
- 2. Dynamic web page:** -- A page has varying (changes) content is known as Dynamic web page.

Types of web Applications:

- **static web application:--** Web application created using static pages is known as static web application.
- **Dynamic web application:--** Web application created using dynamic pages (can also have static pages) is known as Dynamic web application.

=> Static web app is created using HTML mainly.
Ex:- Tutorial sites, Wikipedia sites, few blogs ...etc.
Dynamic web app is created using programming languages Java (Servlet/JSP) .net (ASP.net), PHP,

Architecture of web Applications



Web Server:

A System (Computer) which holds web Applications hosted (web sites) and provides service to web clients (web browser).

Web client:

A System (Computer) which is used to read/access web application by using httpRequest (URL) and gets HttpResponse (Resource).

URL:

Uniform (unique/unified) Resource location (it is a location of one source).

Resource:

Any file (audio/video/text/image/document/html/servlet/jsp) in web site is known as Resource.

Network:

A wireless communication between two(or) more computers is known as network.

Web browser:

A software installed on computer used to make HttpRequest(Acceess web sites).

Ex:-- Google chrome, Firefox, Opera etc..

Web Server:

A System installed on computer to host web sites is known as web servers.

Ex: Apache Tomcat, Apache Httpd etc...

Web Client:

A System has web browser Software is called as Web Client.

Ex:--Mobile phone has Google chrome, Laptop, desktop etc...

End User:

A Person who operates client to make request and gets response.

=> web programmer (Web Designer):- A Person who creates static/dynamic web pages.

Web hosting:

Servlets:

- These are used to develop Dynamic web application also supported static web applications.
- This is an API called as Servlet API [having classes and Interface] provided by sun Micro system.
- This API is provided in 2 Package
 - a. javax.Servlet
 - b. javax.Servlet.http
- Servlet API is provided with basic Interface and Abstract classes along with relations and package.

-----Servlet API-----

javax.Servelt

servlet (I)

IS-A  implements

javax.Servlet

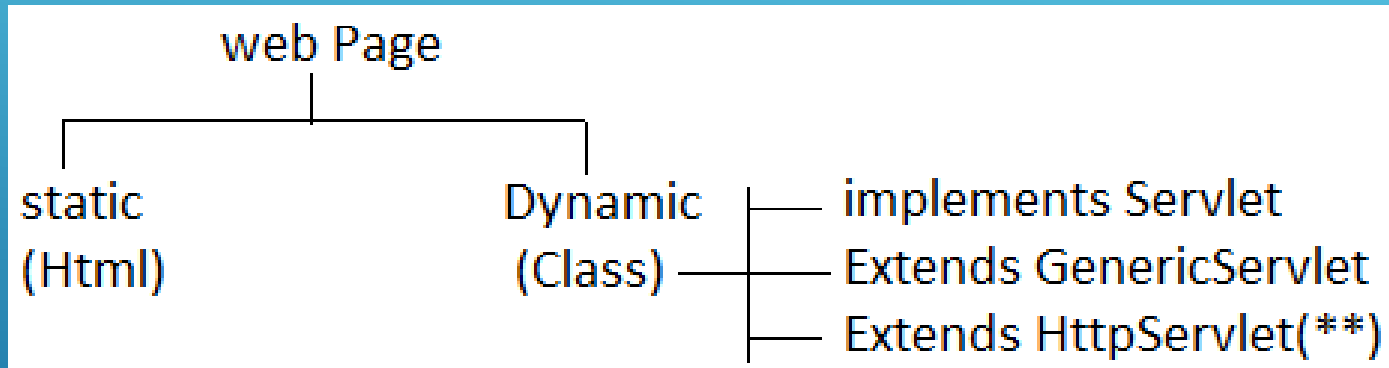
GenericServlet (AC)

HAS-A  Extends

javax.Servlet.http

HttpServlet (A)

=>By using above API we can create one class that provide Dynamic Web page as output.



Web-server

To execute the web resource programs we should use a special software which is called webserver

Webserver executes the web resource programs dynamically when they are requested by clients.

A web server software can also manage, execute multiple web applications simultaneously or parallelly.

Web server software is used to manage and execute web applications

Web container:

A container is a program which provides run time support for other programs

For example a class with main method provides runtime support for other class which does not contain main() method so a main class is container

For example in awt, a frame provides run-time support for Button and labels so a frame is a container.

A web container is a program which provides run time support for servlets and jsp's

Benefits of container:

If there is no container then server should take care of clients request and also servlets and jsp of web applications it is burden on server, so container reduces the burden of server by handling servlets and jsp's

A container manages lifecycle of servlets and jsps .

A container creates a thread for each request and destroys a thread after completion it means container provides multithreading feature

Container restrict users from unauthorized accessing of the servlets and jsp's it means container provides security service.

What is a servlet:

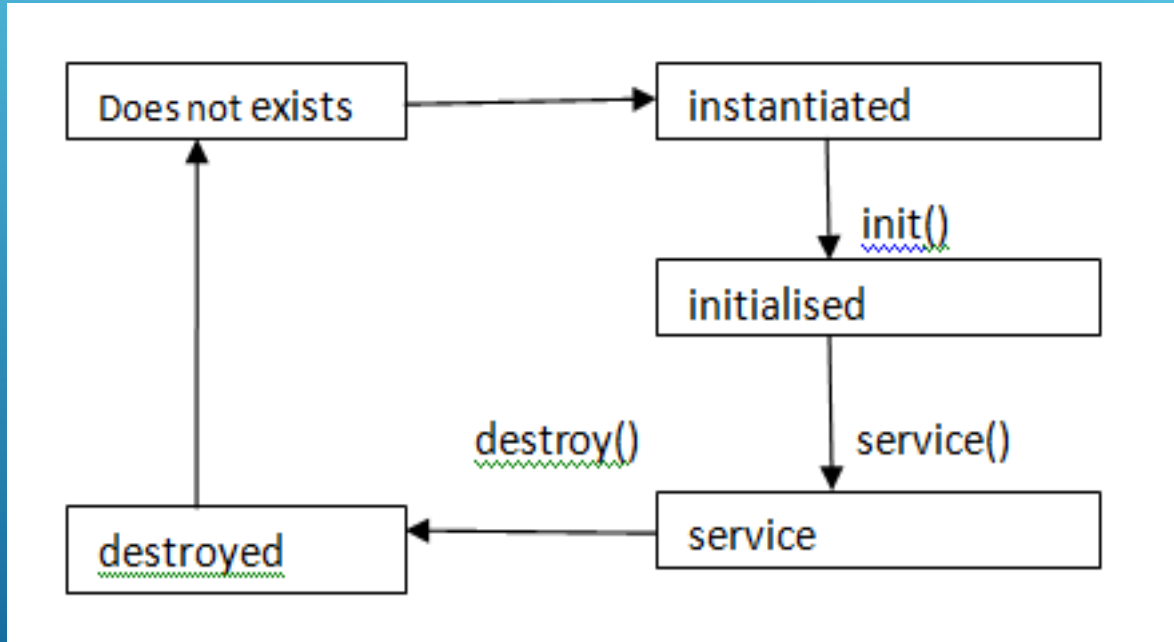
A servlet is a class which runs on a middleware server (webserver or application server), provides its service to multiple clients across internet over http(hyper text transfer protocol)

A servlet is a java program which runs on server and increases the functionality of a server

A servlet is a java program used to accept a client request and process dynamic content as a response

- 1) Servlet is a purely server side program
- 2) Servlets always runs in the server.
- 3) Servlets can be placed in any web and application servers.
- 4) Servlets are useful to develop the dynamic web pages.

Servlet life cycle:



Extending HttpServlet:

HttpServlet is an abstract class and it is a sub class of GenericServlet

HttpServlet has overridden life cycle service() method, which is abstract method in

HttpServlet has all concrete methods only. but still the class is given as Abstract class

In HttpServlet class, we can utilize all the features of Http protocol
actually GenericServlet is a protocol independent servlet so it can handle http protocol also

but we cannot get all the features of http protocol

apart from life cycle service() method in HttpServlet, we have non-life cycle service() method and 7 doXXX() methods

when we are creating a servlet class by extending `HttpServlet` then 3 options are given to the developers to define the logic

1. A developer can override life cycle `service()` method
2. A developer can override non-life cycle `service()` method
3. a developer can override one or more than one `doXXX()` methods

ex:

```
public class Myservlet extends HttpServlet
{
    public void service(ServletRequest request, ServletResponse response) throws
    SE, IOE
    {
        //-----(1)
    }
}
```

```
public class Myservlet extends HttpServlet
{
    public void service(HttpServletRequest request,HttpServletResponse
    response)throws SE,IOE
    {
        //-----(2)
    }
```

```
public class Myservlet extends HttpServlet
{
    public void doPost(HttpServletRequest request,HttpServletResponse
    response)throws SE,IOE
    {
        //-----(3)    }
```

we can differentiate life cycle service() method and non-life cycle service() method based on parameters types, life cycle has ServletRequest, ServletResponse types, But non-life cycle has HttpServletRequest, HttpServletResponse types

In super class(HttpServlet), already life cycle and non-cycle service() method contains some useful logic. if we override then our class will lose the logic defined by super class (HttpServlet)

so we dont use (1) and (2) option

even though 7 doXXX() methods are there but we only use 2 methods, they are doGet(), and doPost()

the 7 doXXX() methods which are available in HttpServlet are

1.doGet()

2.doPost()

if request comes from the browser with get method then in servlet we need to

define the logic in doGet()

if request comes from the browser with post method then in servlet we need to

define the logic in doPost()

some times we define both doGet() and doPost() methods in our servlet . if we want

to execute seperate logic for get and post request then we will override both doGet() and doPost()

doGet() and doPost() are not life cycle methods so they are not directly

the flow is like the following

1.when a request is arrived servlet container calls life cycle service() method

2.life cycle service() method converts ServletRequest , ServletResponse objects in to HttpServletRequest,

HttpServletResponse objects after that it calls non-life cycle service() method

3.Non-life cycle service() method verifies whether a request came from browser with get or post

if get then calls doGet() and if post then calls doPost()

In HttpServlet class, non-life cycle service() method and 7 doXXX() methods are protected methods.

so at the time of overriding, we can use either protected or public modifier

in java when we are overriding , low level access modifier cannot be replaced with high level access modifier. but we can replace high level modifier with the low level modifier

when we dont know whether a request is from browser comes with get or post method

then there are two ways to define a servlet

1.we can write the logic in method(either in doGet() or doPost()) and we can call one method from another method

2.we can write the logic in userdefined method and we can call that userdefined method from doGet() and doPost() methods

Servlet Collaboration/Servlet chaining

In the development of web applications it is not possible to define complete

logic to process a request in a single servlet. in such a case we divide the request

processing logic into multiple servlets.

when logic is developed in multiple servlets, a servlet has to communicate with other servlets to provides a response to a request

for servlets collaboration, servlet technology has given 2 mechanisms

1. dispatching

2. redirection

1. forwarding:

when a request processing logic is divided into multiple servlets then a request

is forwarded from one servlet to another

in forwarding a request processing is finished in destination servlet and response is

also generated in destination servlet.

if any response is generated in source servlet then it will not be considered

when a servlet is forwarding request to another servlet, information will not be given to the client about forwarding.

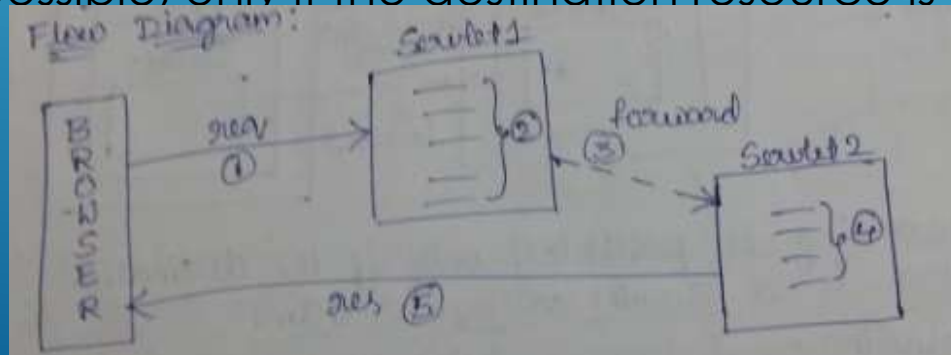
when request forwarded the input values submitted by the client from browser are also automatically forwarded. so, we can read the input values from request object in destination servlet also

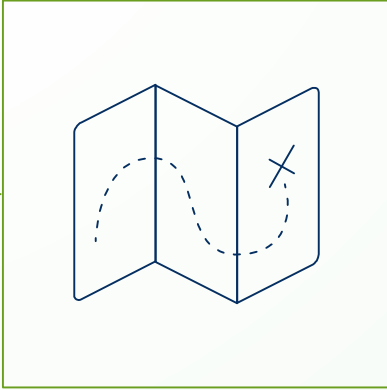
to forward a request both source and destination sources(servlets) must be running

on same server only

If they are in 2 different servers then forwarding is not possible.

forwarding is possible, only if the destination resource is a servlet or jsp page or html.





THANKS!

Any questions?

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